

Lachlan Wright

[lachlanbwwright.github.io \(Portfolio\)](https://lachlanbwwright.github.io) | lachlanbwwright@gmail.com | 0499 469 600

About Me

Full-stack Software Engineer with experience developing and deploying scalable web applications. Looking to deepen my experience and capabilities in building reliable, production-grade systems while continuously expanding my skillset across new frameworks and languages.

Education & Qualifications

Bachelor of Engineering (Honours) (Software) - UTS

Feb 2020 - Nov 2024

- Completed degree with 198 credit points, with a WAM of 85.97%, and a GPA of 6.47/7.00.
- Attained positions on 2022 and 2023 Dean's Lists.

Azure Fundamentals

Jul 2021

- Demonstrated foundational knowledge of cloud services and how they are provided with Azure.

Experience

Software Engineer - Chaos1 (Sydney, NSW)

Nov 2023 - Feb 2026

- Developed a NextJS website with TailwindCSS and ShadCN components for efficient development and consistent interfaces, with a FastAPI backend using SQLAlchemy.
- Contributed to a text editor utilising TipTap's editor, and its live synchronisation with HocusPocus.
- Developed an additional Flutter application with a Spring backend utilising Hibernate.
- Managed Google Cloud Platform services, including Cloud Build, Cloud Run, and Cloud SQL.
- Implemented an end-to-end testing suite executed using Cypress and GitHub Actions.
- Provided team members with technical mentoring, guidance, and code reviews.

Programmer (Freelancer) - DataAnnotation (Remote)

Jul 2024 - Present

- Wrote, reviewed, tested, and adjusted programming prompts and their responses to assist the development of Large Language Models.

Portfolio

Pangea Resource Edit

- Formulated a drag and drop level editor for games from Pangea Software with React and Konva.
- Forked and rewrote the Python "RSRCDump" library for editing MacOS Resource Fork files in TypeScript and extended it to support padding bytes, and improve support for struct arrays.
- Analysed the games' source code to understand and interpret level data structures.
- Implemented the LZSS compression algorithm for decompressing and repacking level data.
- Developed converter for translating model data from proprietary "BG3D" and depreciated "3DMF" (QuickDraw 3D) formats with separate skeletal and animation data to glTF and back.
- Used ThreeJS via React Three Fiber to visualise model data and level topology.

Online Mafia Game

- Constructed an online chat-based game built with NextJS, in addition to a React Native mobile app.
- Handled bidirectional user-server WebSocket communication using SocketIO and PartyKit.

Driving Game and Level Editor

- Ported a Mac OS 9 to WebAssembly using existing research and agentic development.
- Produced Angular-based level and asset editors hosted alongside the game on GitHub Pages using Angular Material and Konva Libraries.

Public Transport Timetable

- Developed a Flutter mobile application for viewing public transportation timetables in NSW.
- Interfaced with Transport for NSW's OpenData API, parsing Protobuf and CSV data.
- Persisted local data in a database with Drift, and displayed map information with Flutter Map.

Deal Scanner

- Devised a TypeScript application for monitoring online retailers to find and notify users of deals, deployed using Heroku alongside a database hosted using Turso.
- Queried endpoints, scraped websites with Puppeteer, and sent notifications leveraging DiscordJS.
- Utilised an SQLite database via Prisma to persist queries created by users.

Online Exam System

- Created a mock website for online invigilation of exams with NextJS, tRPC, Prisma, and TailwindCSS.
- Provided advice, feedback, guidance, and performed code reviews in an 8-person team.
- Contributed to an implementation of a system for blurring the background of a webcam feed.

Cryogenic Wiring Simulation

- Developed a ReactJS website for visually interfacing with a Python model simulating how a cryogenic computer's wiring configuration influences its temperature and performance.
- Visualised model outputs using Recharts.
- Communicated with clients to clarify specifications and demonstrate progress.

Technical & Core Capabilities

Languages: C, C#, CSS, Dart, HTML, Java, JavaScript, Kotlin, MATLAB, Python, Rust, TypeScript

Frameworks: Angular, Flutter, NextJS, ReactJS, Spring, Svelte, SvelteKit, Vue

Libraries: Bootstrap, ExpressJS, Immer, Jotai, PartyKit, Pyodide, ShadCN, SocketIO, TailwindCSS, TanStack, TipTap, tRPC

Testing, Deployment: Azure, Docker, GitHub Actions, GitLab CI/CD, Google Cloud Platform, Heroku, Pytest, Vitest

Database Management: Hibernate, MongoDB, Mongoose, PostgreSQL, Prisma, SQLAlchemy, SQLite

Data Science: KNIME, MATLAB, Numpy, Pandas, Tensorflow

Agentic Tools: Antigravity, Claude Code, Codex, Gemini CLI, GitHub Copilot, OpenCode